# Managerial Economics

## Lecture 5 Cost Analysis

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# Roadmap

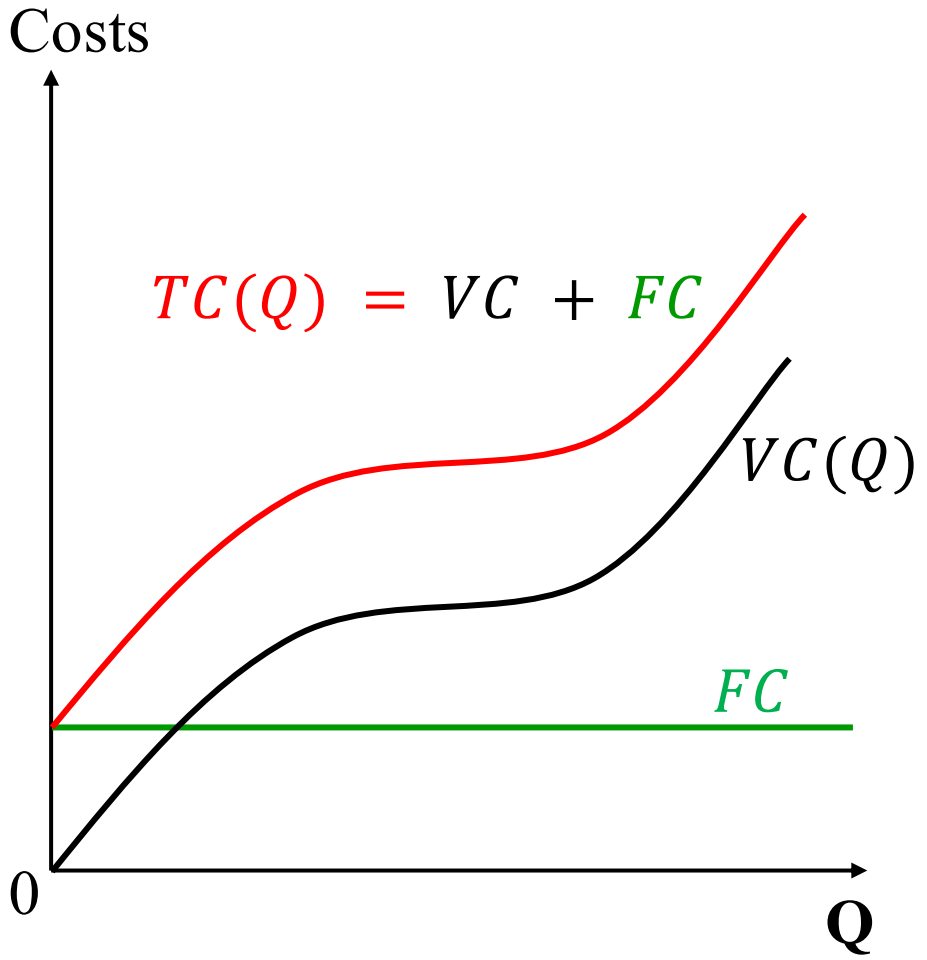
- Types of Costs in Managerial Decisions
  - **Short-Run:** Fixed costs, Sunk costs, Variable costs, Total costs
  - **Long-Run:** All costs are variable, no fixed costs

- Cost-reducing strategies:
  - Learning-by-doing
  - Economies of Scope

# Different Types of  Short-Run Costs

# Fixed , Variable Costs and Total Costs

- **Fixed costs (*FC*，固定成本)** do not vary with the quantity of output produced.
  - *E.g.*, *FC* = $10000 for renting the railcar
  - *Other examples:* cost of equipment, loan payments, rents paid for a whole year
- **Variable costs (*VC*，可变成本)** vary with the quantity produced.
  - E.g. , *VC* = wages he pays workers
  - *Other example:* cost of materials
- **Total cost (*TC*，总成本) = *FC* + *VC***



# More Production Costs Definitions

- **Average Total Cost (ATC, 平均总成本)** is the sum of **Average Variable Cost (AVC, 平均可变成本)** and **Average Fixed Cost (AFC, 平均固定成本)**

$$ATC = AVC + AFC$$
$$ATC = TC(Q)/Q$$

- **Average Variable Cost** is the variable cost of producing one unit of output

$$AVC = VC(Q)/Q$$

- **Average Fixed Cost** is the fixed cost of producing one unit of output

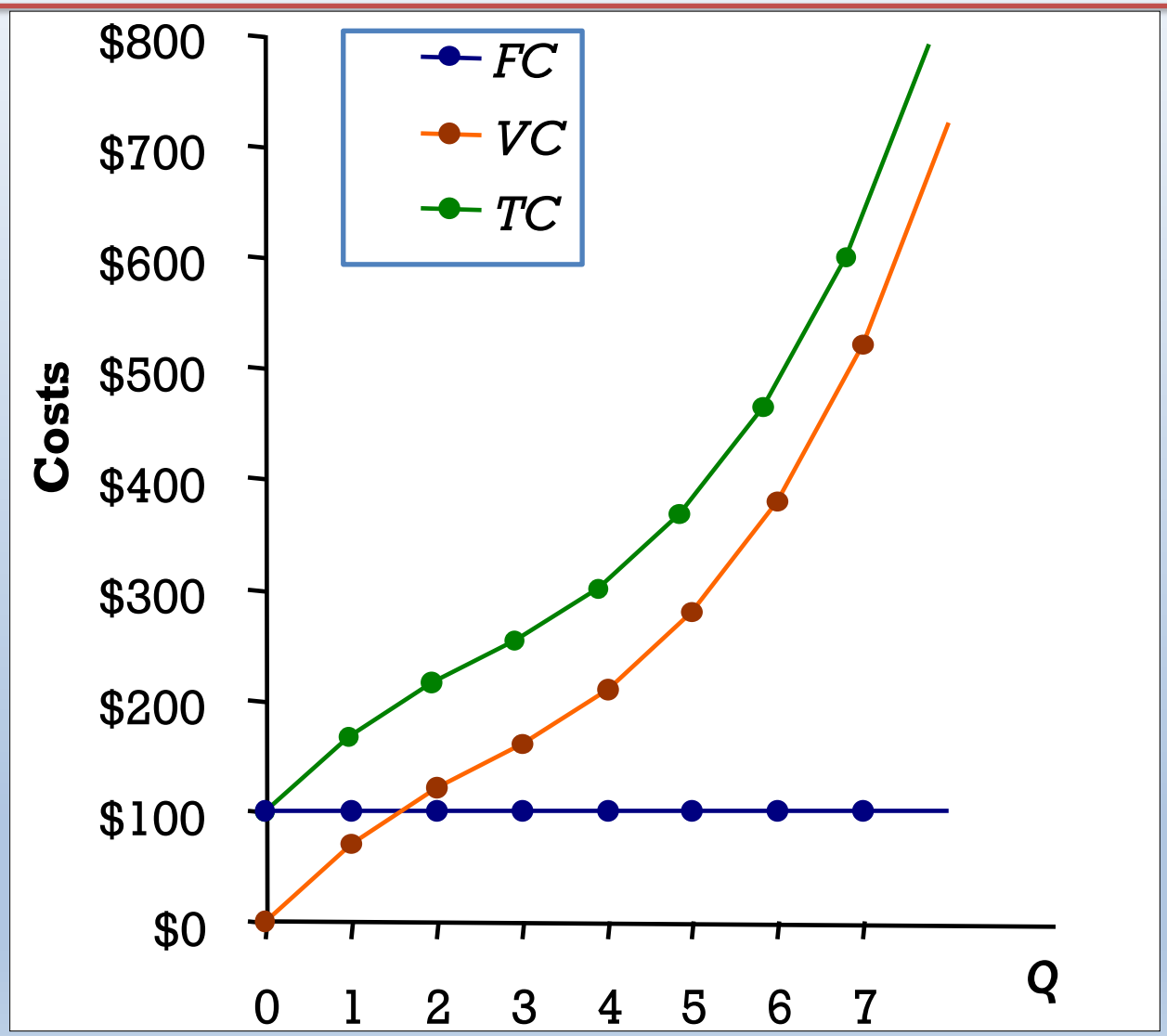
$$AFC = FC/Q$$

- **Marginal Cost** is the change in total cost for a marginal change in the output:

$$MC = \Delta TC/\Delta Q$$

# EXAMPLE: Fixed Cost, Variable Cost and Total Cost

| Q | FC | VC | TC |
|---|---|---|---|
| 0 | $100 | $0 | $100 |
| 1 | 100 | 70 | 170 |
| 2 | 100 | 120 | 220 |
| 3 | 100 | 160 | 260 |
| 4 | 100 | 210 | 310 |
| 5 | 100 | 280 | 380 |
| 6 | 100 | 380 | 480 |
| 7 | 100 | 520 | 620 |

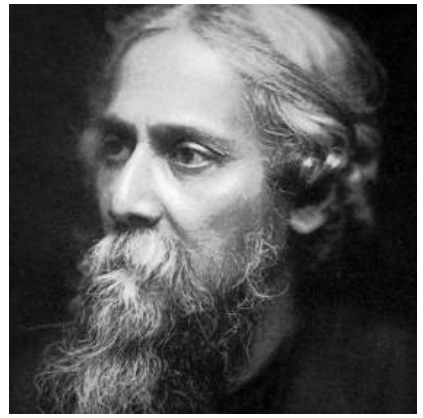


# Fixed and Sunk Costs

- **Fixed Cost:** Costs that do not change as output changes.

- **Sunk Cost (沉没成本):** A cost that is forever lost after it has been paid.
- In real-life cases, sunk cost is *usually a part of the fixed cost*
- **Decision makers should ignore sunk cost to maximize profit or minimize losses**

**Don't Cry over the Spilt Milk!**



Rabindranath Tagore (泰戈尔), India Poet, 1861-1941

# Quiz: Sunk Costs in Decision-Making

Bob is the manager of a coal company and has just paid $10,000 to lease a railcar for one month to ship the coal to the market. Suppose that after signing the lease agreement, there is *a sudden drop in market demand and the coal company no longer needs the railcar*. Another company proposes to sublease the railcar using $2,000.

**Managerial Question:** If the original lease agreement allows Bob to sublease the railcar, should Bob sublease the railcar?

# EXAMPLE 1: Farmer Bob's Costs

- Farmer Bob must pay $1,000 per month for the land, regardless of how much wheat he grows.
- The market wage for a farm worker is $2,000 per month.
- So farmer Bob's costs are related to how much wheat he produces….

- The laws behind this example are general!

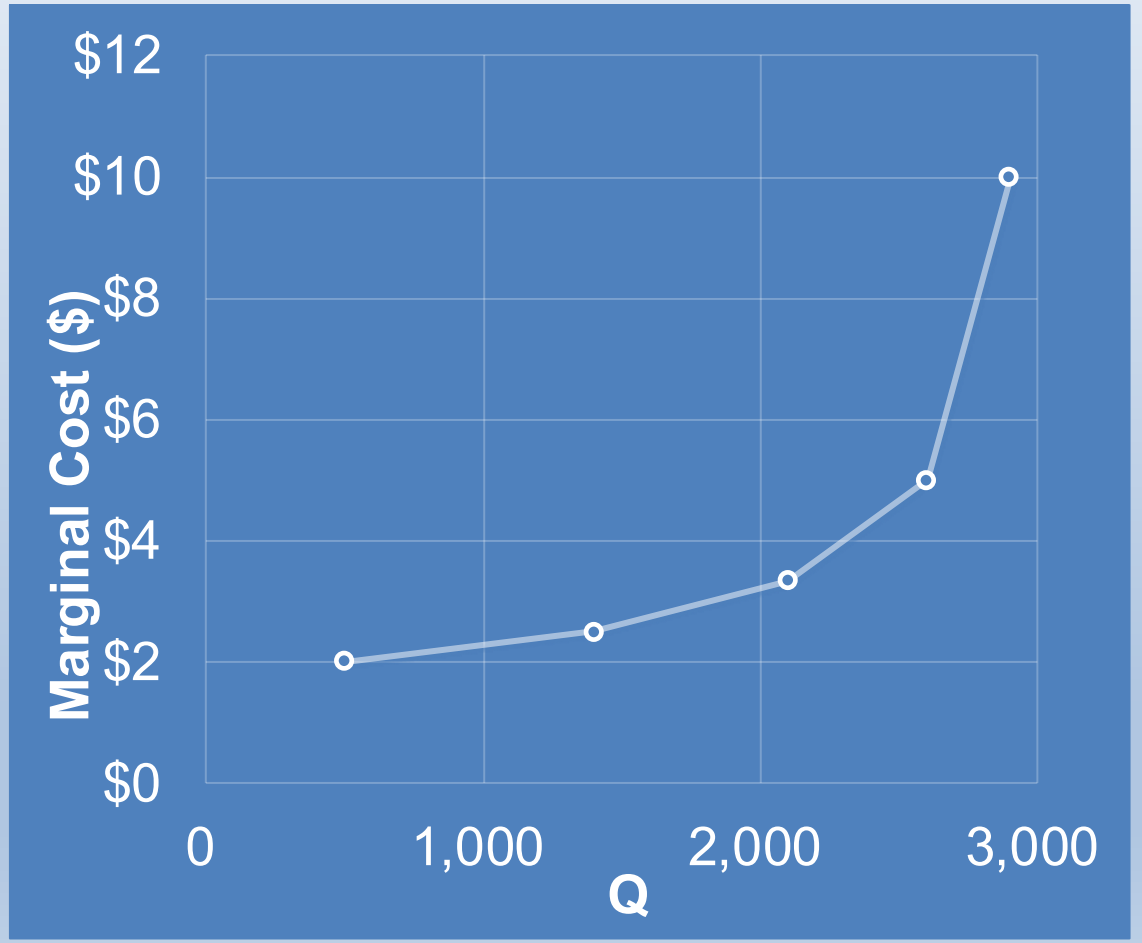
# Total and Marginal Cost

| | Q (kg of wheat) | Total Cost | | Marginal Cost (*MC*) |
|---|---|---|---|---|
| | 0 | $1,000 | | |
| $\Delta Q = 1000$ | | | $\Delta TC = \$2000$ | $2.00 |
| | 1000 | $3,000 | | |
| $\Delta Q = 800$ | | | $\Delta TC = \$2000$ | $2.50 |
| | 1800 | $5,000 | | |
| $\Delta Q = 600$ | | | $\Delta TC = \$2000$ | $3.33 |
| | 2400 | $7,000 | | |
| $\Delta Q = 400$ | | | $\Delta TC = \$2000$ | $5.00 |
| | 2800 | $9,000 | | |
| $\Delta Q = 200$ | | | $\Delta TC = \$2000$ | $10.00 |
| | 3000 | $11,000 | | |

# The Marginal Cost Curve

| *Q* (kg of wheat) | *TC* | *MC* |
|---|---|---|
| 0 | $1,000 | |
| | | $2.00 |
| 1000 | $3,000 | |
| | | $2.50 |
| 1800 | $5,000 | |
| | | $3.33 |
| 2400 | $7,000 | |
| | | $5.00 |
| 2800 | $9,000 | |
| | | $10.00 |
| 3000 | $11,000 | |

**MC** usually rises as ***Q*** rises, **Why?**

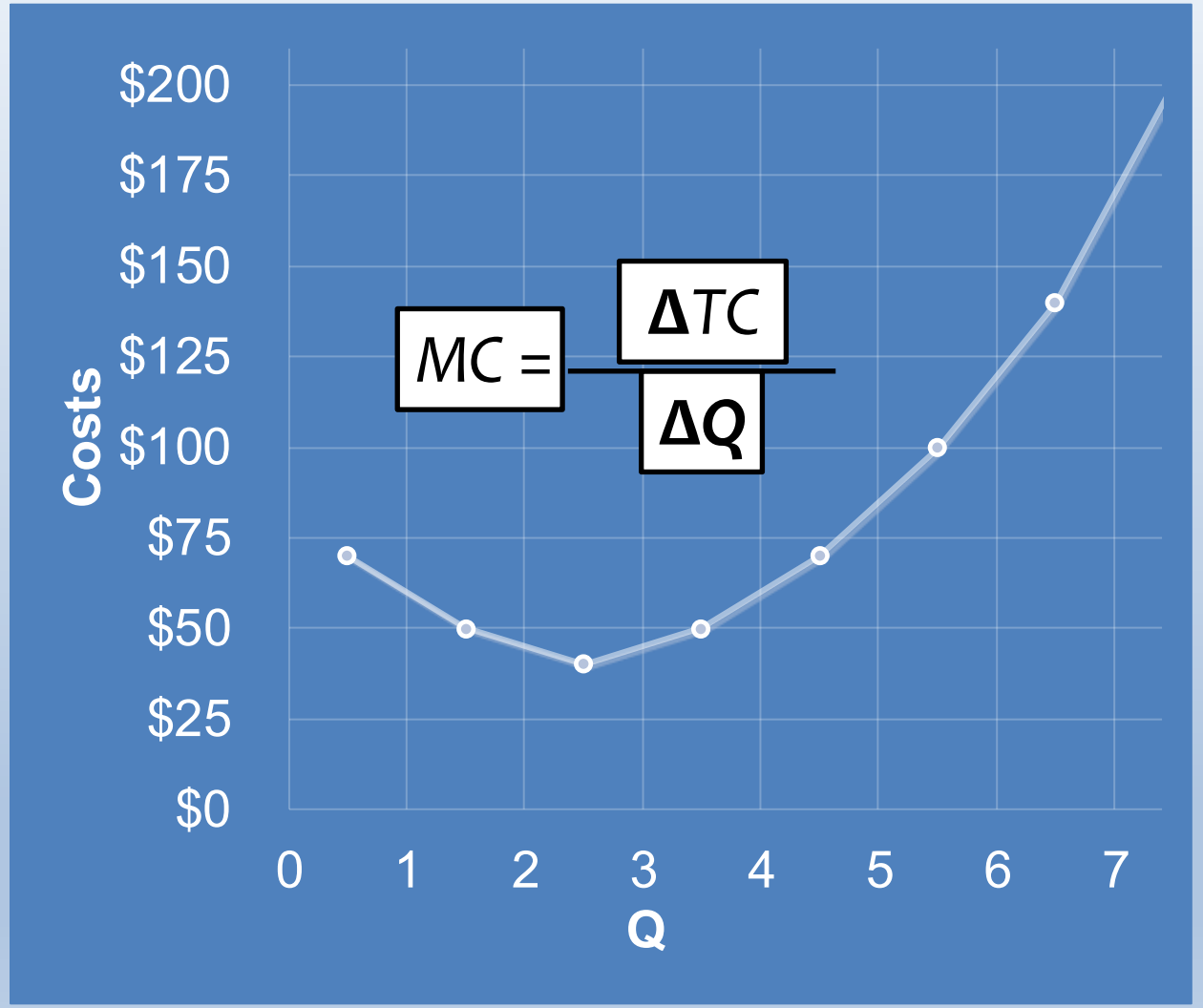


# Why Does MC Rise as Output Rises?

- Marginal product normally exhibits diminishing returns.
- When *the marginal product of an extra unit of input decreases*, it also implies that
  - The marginal input for an extra unit of output increases
  - The marginal cost of an extra unit of output increases
- Diminishing MP and increasing MC are two sides of the same coin.
- Nonetheless, at the very early stage of production
  - Possible for **MP to increase:** due to better utilization of fixed inputs, e.g., machines
  - **MC decreases at this stage**

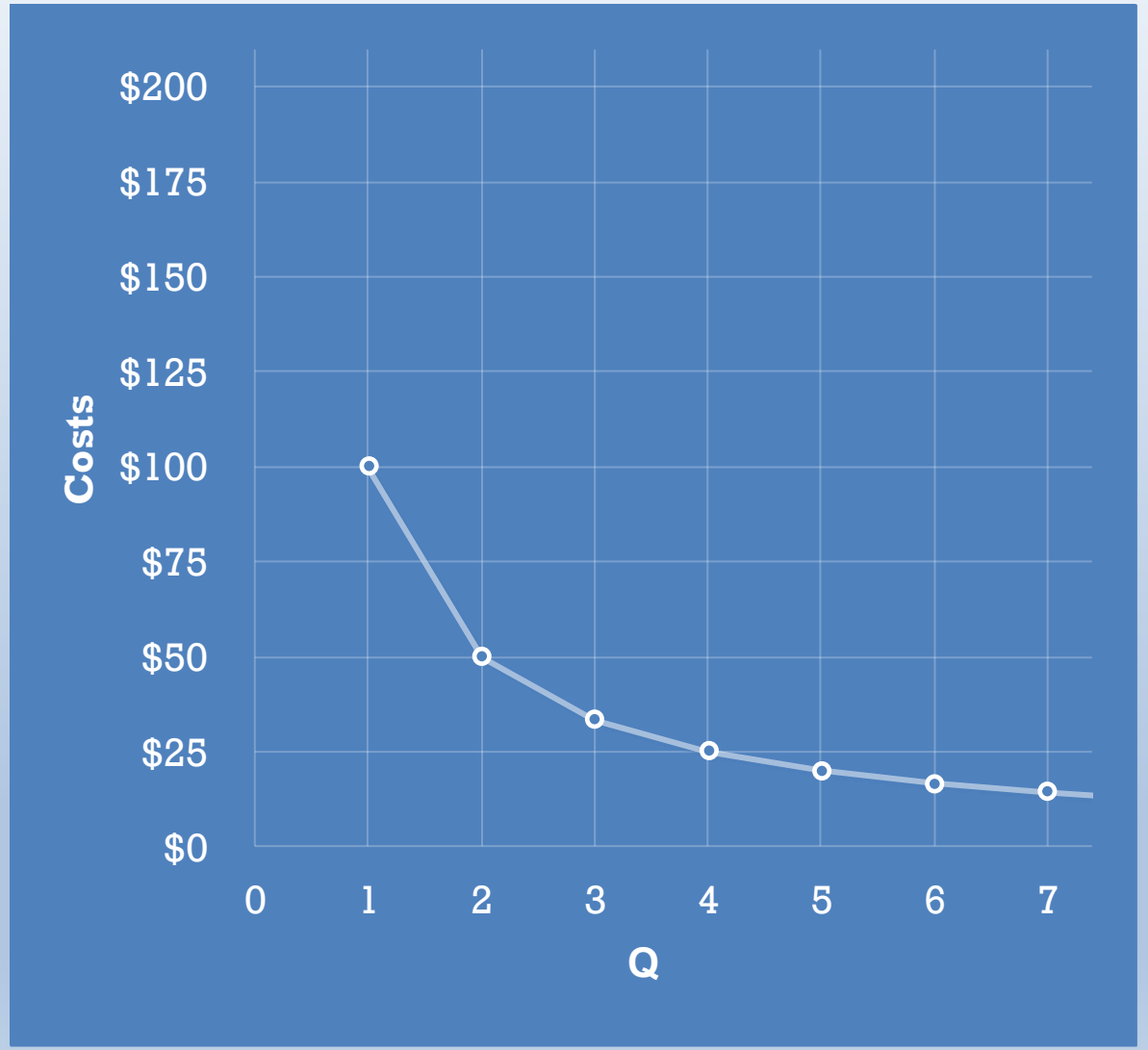
# Marginal Cost Curve with Initial Production Stage

| Q | TC | MC |
|---|---|---|
| 0 | $100 | |
| | | $70 |
| 1 | 170 | |
| | | 50 |
| 2 | 220 | |
| | | 40 |
| 3 | 260 | |
| | | 50 |
| 4 | 310 | |
| | | 70 |
| 5 | 380 | |
| | | 100 |
| 6 | 480 | |
| | | 140 |
| 7 | 620 | |



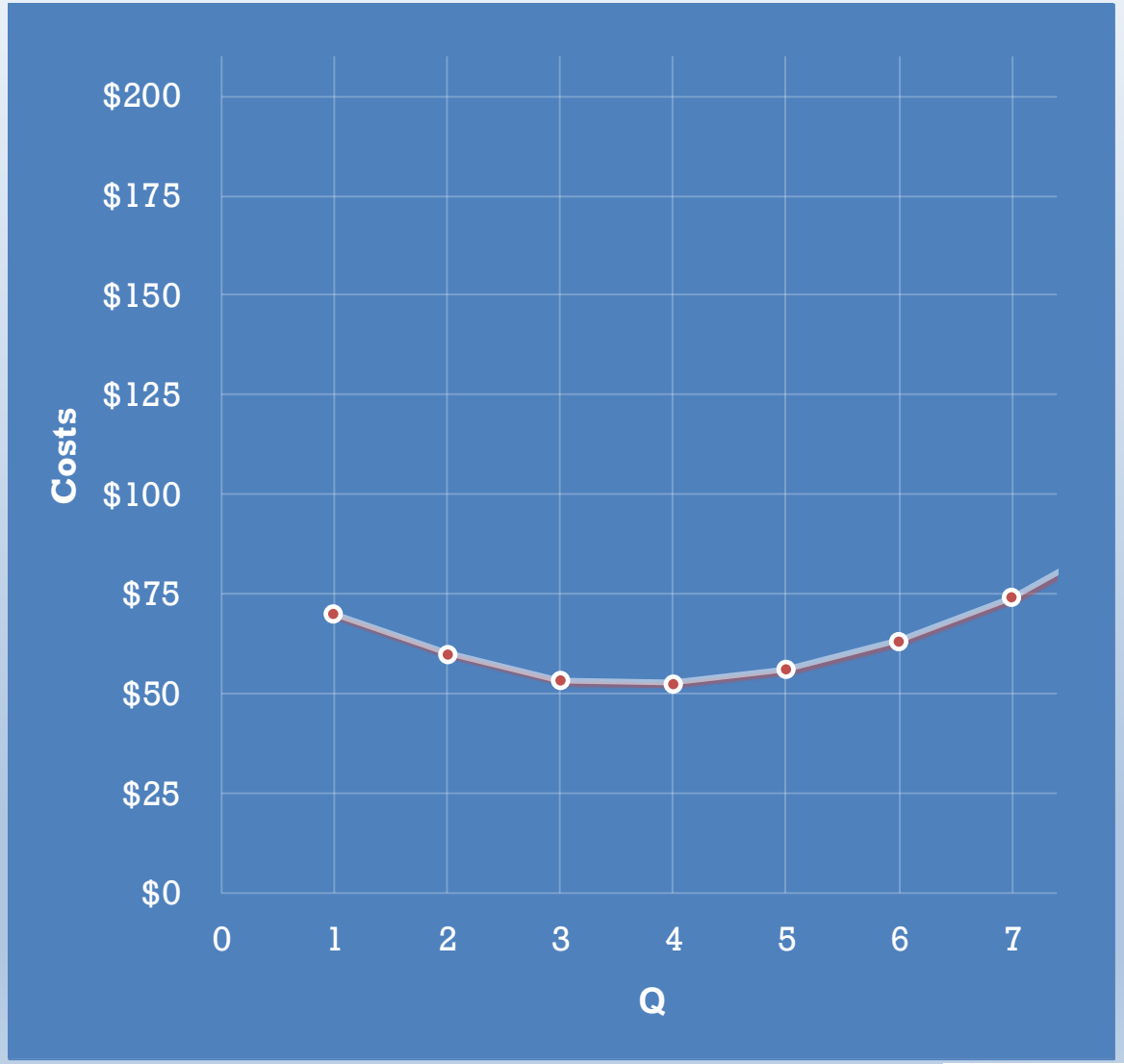
# Average Fixed Cost

| Q | FC | AFC |
|---|---|---|
| 0 | $100 | n/a |
| 1 | 100 | $100 |
| 2 | 100 | 50 |
| 3 | 100 | 33.33 |
| 4 | 100 | 25 |
| 5 | 100 | 20 |
| 6 | 100 | 16.67 |
| 7 | 100 | 14.29 |



# Average Variable Cost

| Q | VC | AVC |
|---|---|---|
| 0 | $0 | n/a |
| 1 | 70 | $70 |
| 2 | 120 | 60 |
| 3 | 160 | 53.33 |
| 4 | 210 | 52.50 |
| 5 | 280 | 56.00 |
| 6 | 380 | 63.33 |
| 7 | 520 | 74.29 |



# Discussion: Analyze Relevant Costs in a Managerial Decision

Bob is the manager of a coal company and has just paid $10,000 to lease a railcar for one month to ship the coal to the market. In the managerial decision, what are the relevant types of costs?

- ✓ **Fixed Costs (固定成本):** As mentioned, the $10,000 railcar lease payment is a fixed cost, *as it does not change regardless of the production volume.*

- ✓ **Variable Costs (可变成本):** *As the production volume increases, the variable costs of the coal company also increase.* For example, the costs of mining and transporting coal rise with the amount of coal that needs to be transported.

- ✓ **Sunk Costs (沉没成本):** Suppose that after signing the lease agreement, there is a sudden drop in market demand and the coal company no longer needs the railcar. *The already paid lease payment cannot be recovered, even if the railcar is not used, making it a sunk cost.*

- ✓ **Marginal Costs (边际成本):** The *additional cost incurred for producing and transporting each additional ton of coal* is the marginal cost. For instance, producing an extra ton of coal might mean additional fuel and labor costs.

# Average Total Cost

| Q | TC | ATC | AFC | AVC |
|---|---|---|---|---|
| 0 | $100 | n/a | n/a | n/a |
| 1 | 170 | $170 | $100 | $70 |
| 2 | 220 | 110 | 50 | 60 |
| 3 | 260 | 86.67 | 33.33 | 53.33 |
| 4 | 310 | 77.50 | 25 | 52.50 |
| 5 | 380 | 76 | 20 | 56.00 |
| 6 | 480 | 80 | 16.67 | 63.33 |
| 7 | 620 | 88.57 | 14.29 | 74.29 |

**Average total cost (*ATC*,平均总成本)** equals total cost divided by the quantity of output:

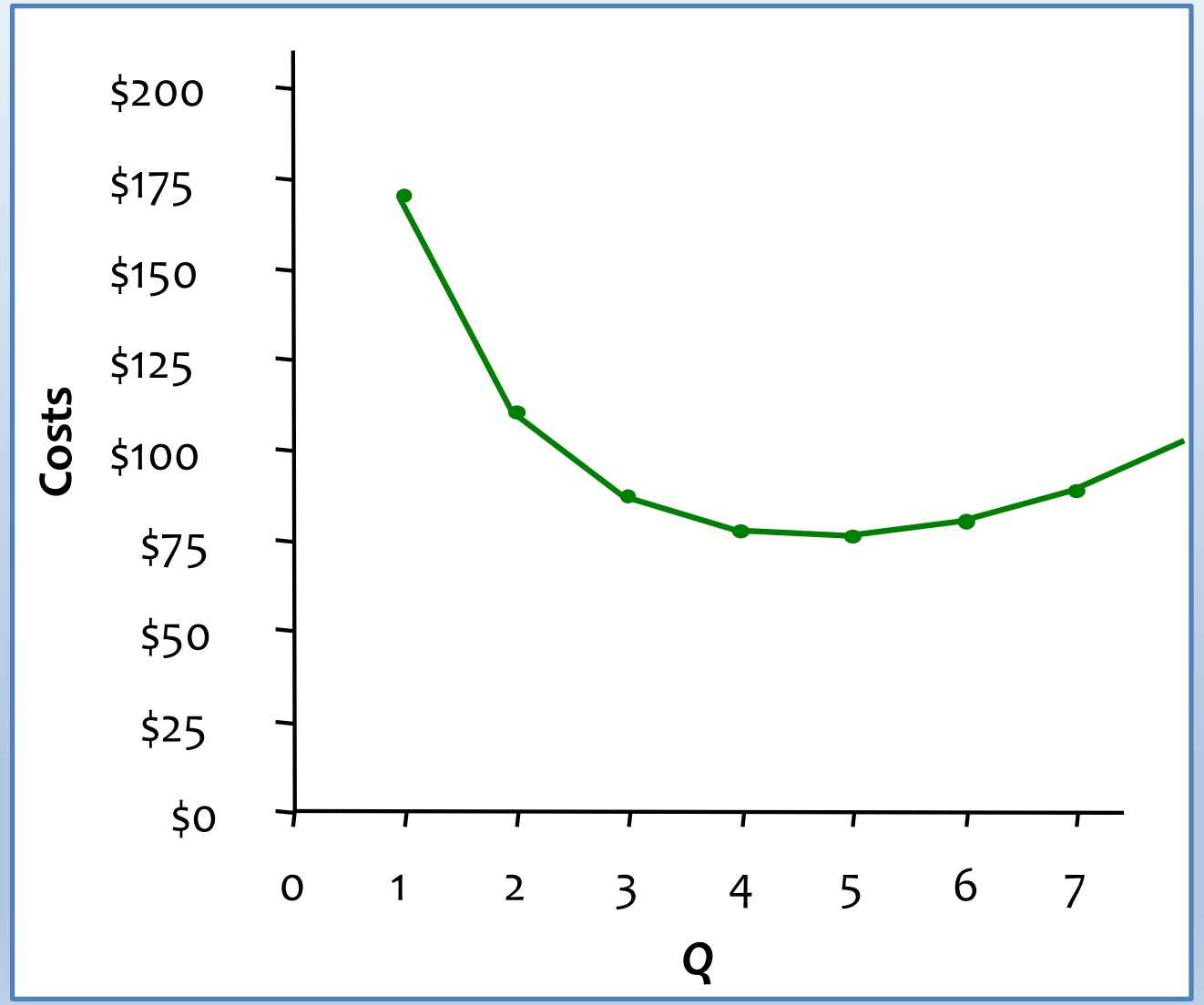
$$ATC = TC/Q$$

Also,

$$ATC = AFC + AVC$$

# Average Total Cost Curve

| Q | TC | ATC |
|---|---|---|
| 0 | $100 | n/a |
| 1 | 170 | $170 |
| 2 | 220 | 110 |
| 3 | 260 | 86.67 |
| 4 | 310 | 77.50 |
| 5 | 380 | 76 |
| 6 | 480 | 80 |
| 7 | 620 | 88.57 |



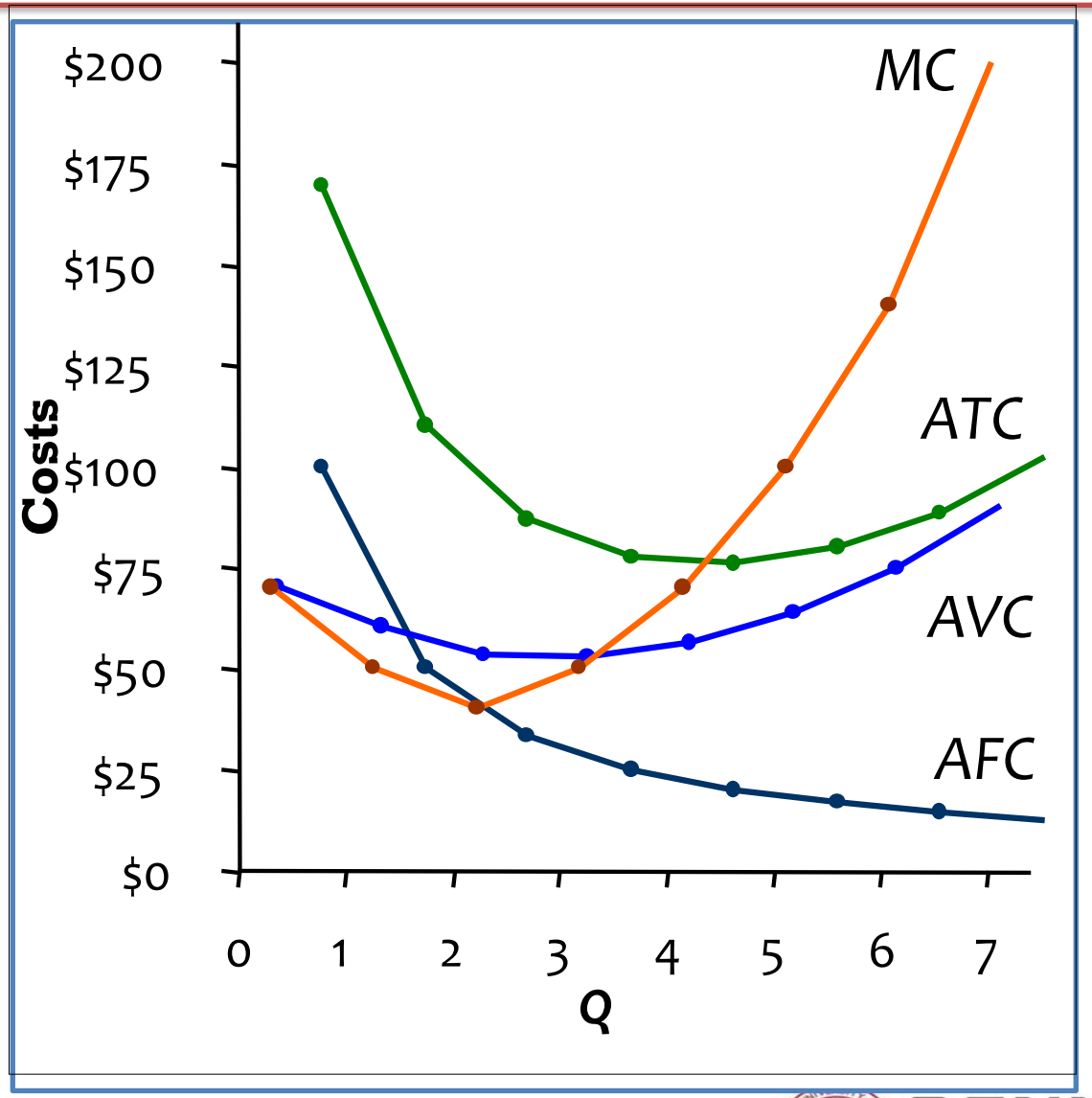
# Relationships Between Different Short-run Costs

- **What is the relationship between MC and AVC?**
  - MC measures the marginal contribution of an extra unit of output to total cost
  - **MC < AVC:** Marginal contribution < average contribution of existing cost. AVC is pulled down;
  - **MC>AVC:** AVC is lifted up by the higher MC

***The MC curve crosses the AVC curve at the AVC curve's minimum.***

- **What is the relationship between MC and ATC?**

***The MC curve goes across the lowest point of ATC***





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# Practice: Calculating Costs

Fill in the blank cells of this table.

| Q | VC | TC | AFC | AVC | ATC | MC |
|---|---|---|---|---|---|---|
| 0 | | $50 | | | | |
| 1 | 10 | | | $10 | $60.00 | $10 |
| 2 | 30 | 80 | | | | |
| 3 | | | 16.67 | 20 | 36.67 | 30 |
| 4 | 100 | 150 | 12.50 | | 37.50 | |
| 5 | 150 | | | 30 | | |
| 6 | 210 | 260 | 8.33 | 35 | 43.33 | 60 |

# Quiz: Cost Function and Costs

- Suppose we have the following information:

$$TC = 200 + 5Q - 0.4Q^2 + 0.001Q^3$$

$$MC = 5 - 0.8Q + 0.003Q^2$$

1. Find the fixed cost

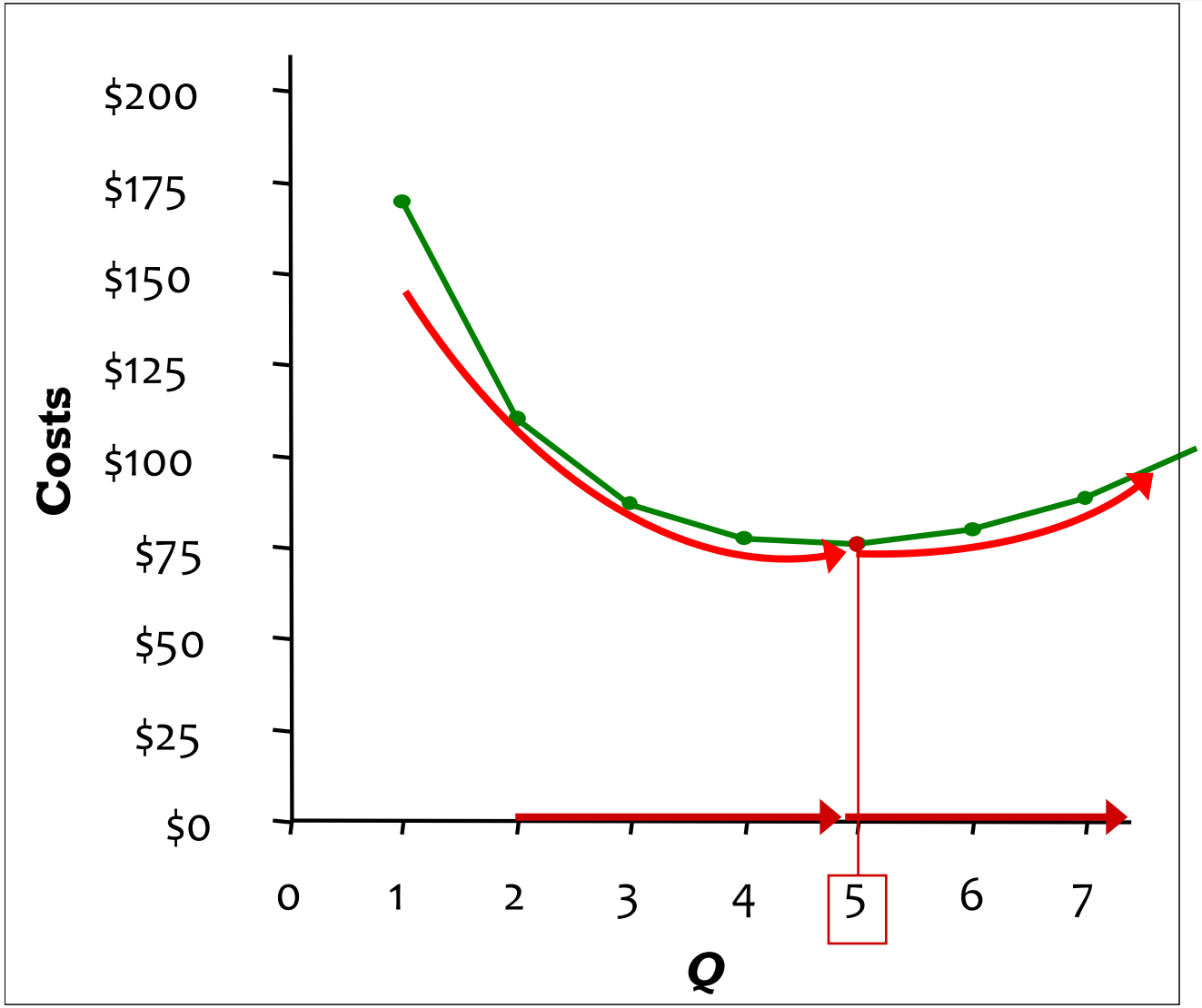
2. Find AVC function

3. Find AVC at Q = 10

4. If FC rises $500, what happens to the average variable cost function?

# Efficient Scale

- As ***Q*** rises:
  - Initially, falling *AFC* pulls *ATC* down.
  - Eventually, rising *AVC* pulls *ATC* up.
- **Efficient scale (效率规模)**:
  The quantity that minimizes ATC.



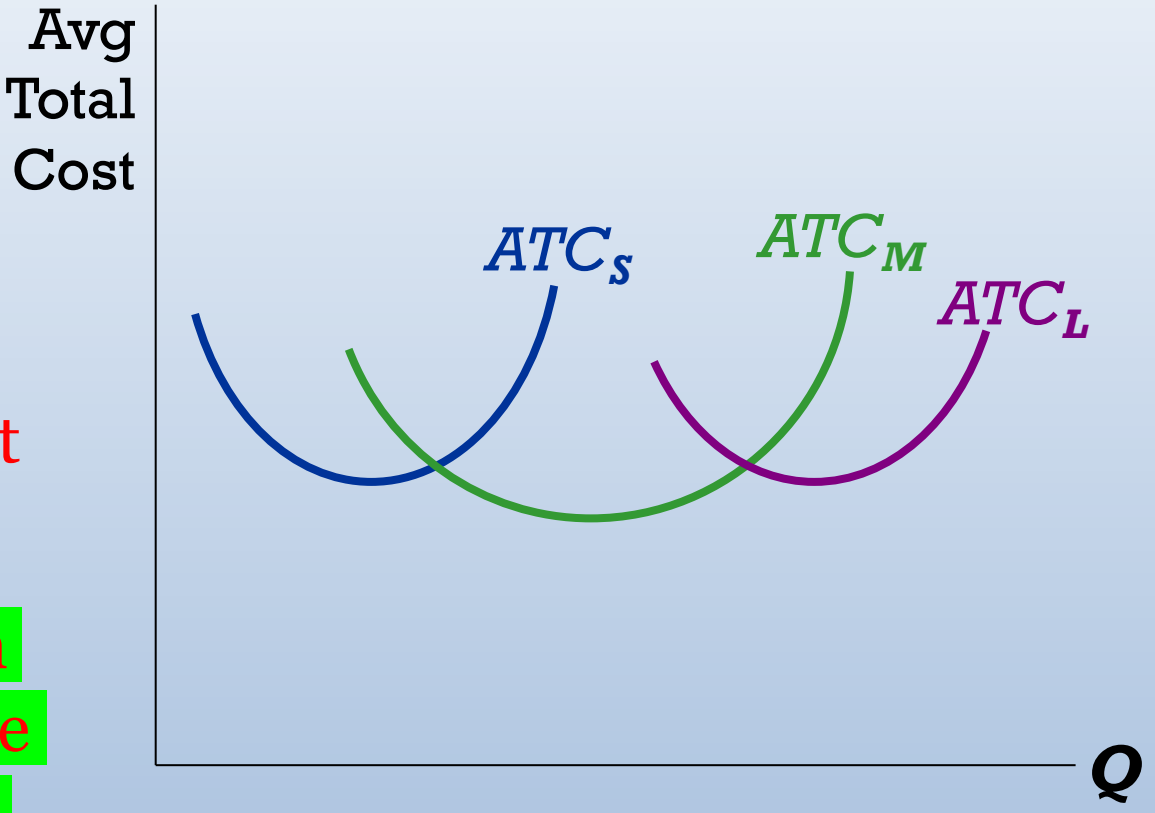
# Long-Run Costs

# Costs in the Long Run

- **Short run:**
  - Some inputs are fixed (*e.g.*, factories, land). The costs of these inputs are *FC*.
- **Long run:**
  - All inputs are variable (*e.g.*, firms can build new factories, or sell old ones).
- In the long run, *ATC* at any ***Q*** is cost per unit using the most efficient mix of inputs for that ***Q*** (*e.g.*, the factory size with the lowest *ATC*).

# EXAMPLE 2: Long-run ATC with 3 factory sizes

- Firm can choose from 3 factory sizes: **S**, **M**, **L**.
- Each size has its own *short-run ATC* curve.
- Why different positions represent different sizes?



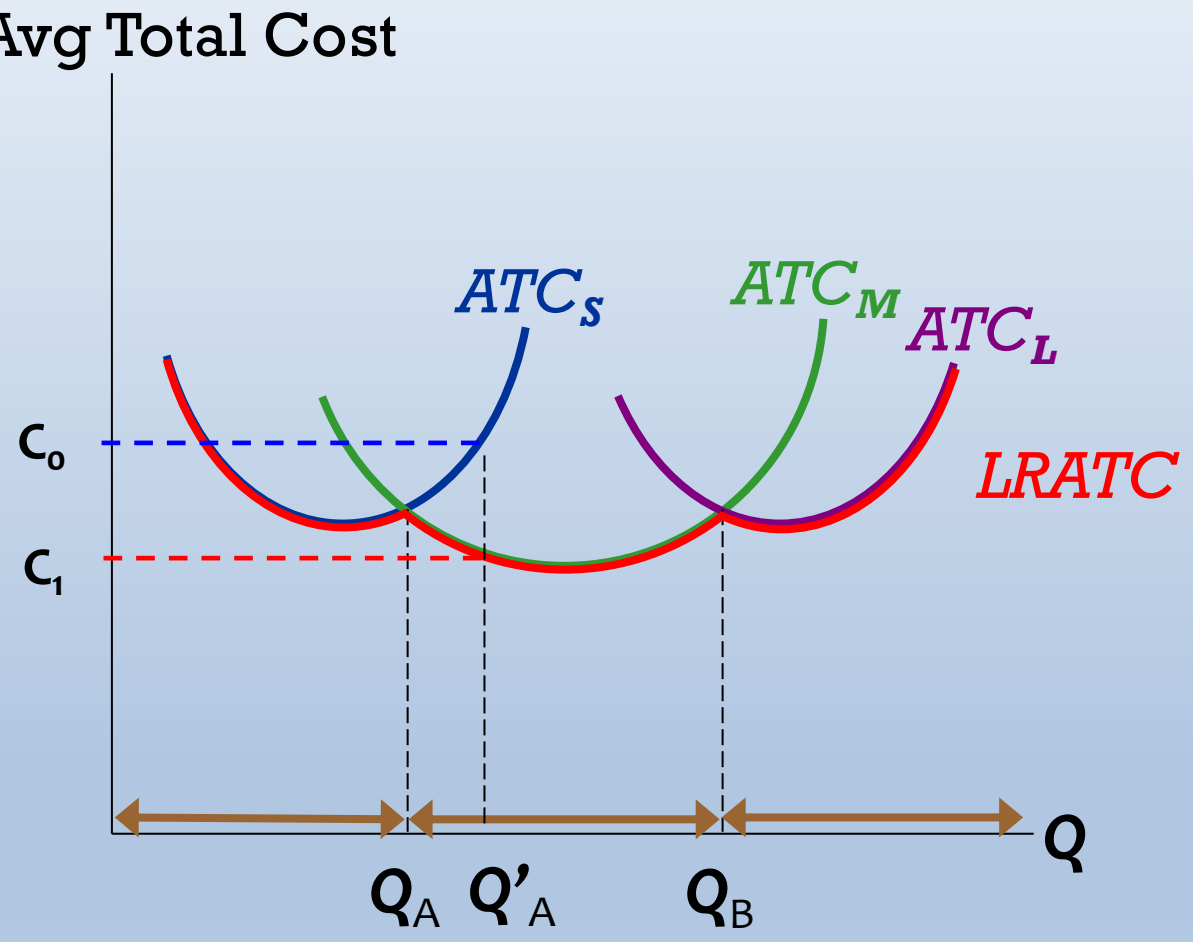

Recall ATC=AVC+(FC/Q). The firm can change to a different factory size in the long run, but not in the short run.



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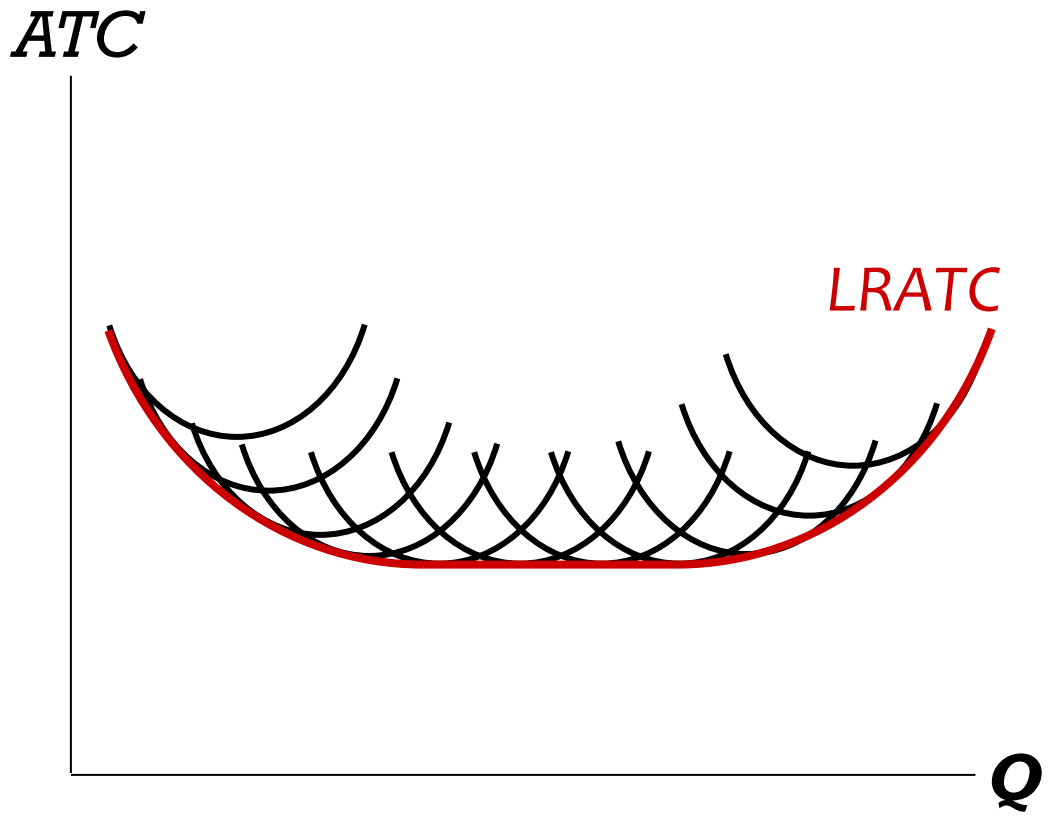
# EXAMPLE 2: Long-run ATC with 3 Factory Sizes

- **In the long run**, to produce less than $Q_A$, firm will choose size **S**.
  - Assume an urgent order comes ($Q'_A$), cost goes up to $C_o$ in the short run
  - But the firm can expand to size M, so that cost goes down to *C1*
- **In the long run,** to produce between $Q_A$ and $Q_B$, firm will choose size **M**
- **In the long run,** to produce more than $Q_B$, firm will choose size **L**



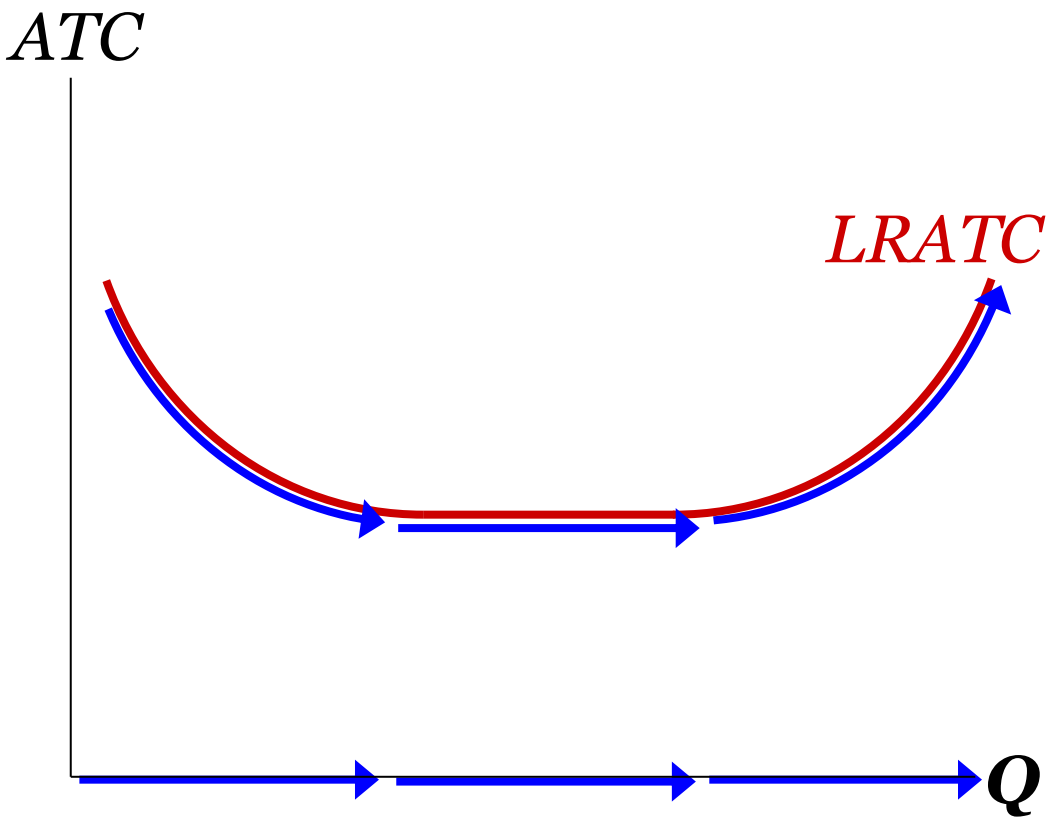
# A Typical Long-run ATC Curve

- In long run, the firm can choose any level of capacity
- Once it commits to a level of capacity, at least one of the inputs must be fixed. This then becomes a short-run scenario
- The LR ATC curve is an envelope（包络线="信封"）of SR ATC curves



# Long-run Production Cost Curve

- **Economies of scale（规模经济）**: *ATC* falls as ***Q*** increases.
- **Constant returns to scale（规模收益不变）**: *ATC* stays the same as ***Q*** increases.
- **Diseconomies of scale（规模不经济）**: *ATC* rises as ***Q*** increases.



# Scale of Production Long-run ATC

Reasons for long-run *Economies of Scale:*

- Specialization of labor and capital
  - ✓ Workers are more efficient when focusing on a narrow task
- Prices of raw materials may fall with volume discounts in firm's purchasing
  - ✓ Stronger bargaining power
- Larger firms may be able to raise funds in capital markets at a lower cost
  - ✓ Small-sized enterprises are hard to get bank loans, but mega-sized private enterprises & SOEs in China can easily get long-term loans from commercial banks
- Larger firms may be able to spread out *promotional costs*, e.g.
  - ✓ Advertising budget might not be changed if a firm's scale of production expands by 10%

# Economies of Scale: Colgate's Global Strategy

- **Colgate** sells standard toothpaste in more than 40 countries, utilizing the economies of scale for the exterior design:
  - **Unified Packaging Design:** Reduces the need for designing different packages for different markets, thereby decreasing the complexity of design and production and saving millions of dollars.
  - **Reduced Marketing Costs:** Reduces the need for customizing advertising and marketing materials for each market, saving marketing and advertising expenses.
  - **Advantages of Bulk Purchasing:** Unified packaging design allows the company to purchase packaging materials in bulk, thus gaining cost advantages in procurement.
  - **Enhanced Brand Recognition:** Consumers seeing similar packaging in different countries can more easily recognize and trust the brand.

# The Popularity of Economies of Scale

- Economies of scale has a strong explanatory power: Maximizing resource utilization

Cloud services

阿里云 aliyun.com

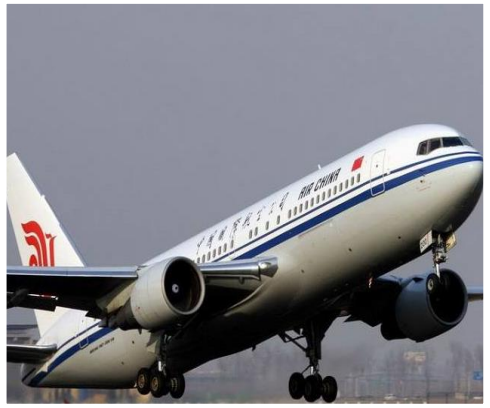
24/7 oil refinery



Sharing economy

滴滴出行
滴滴一下 美好出行

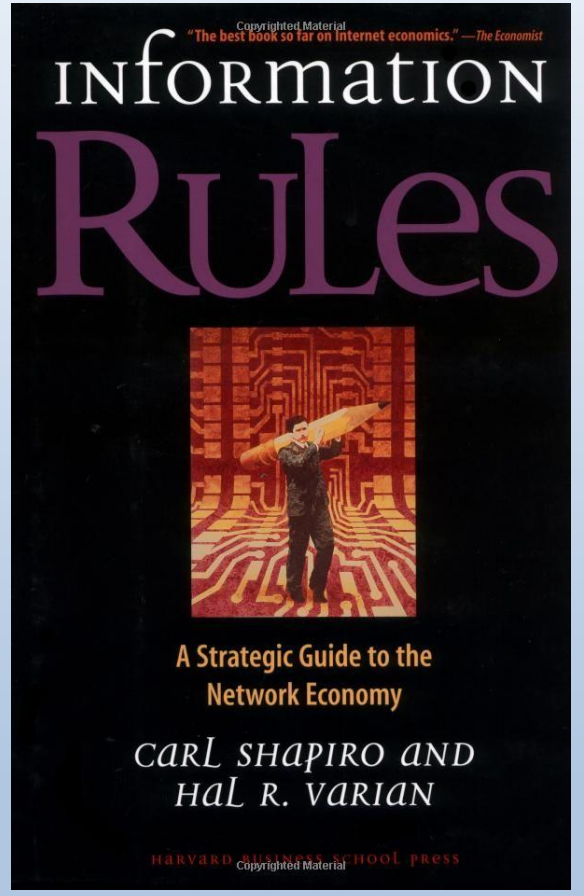
Airlines



# The Cost of Producing Information

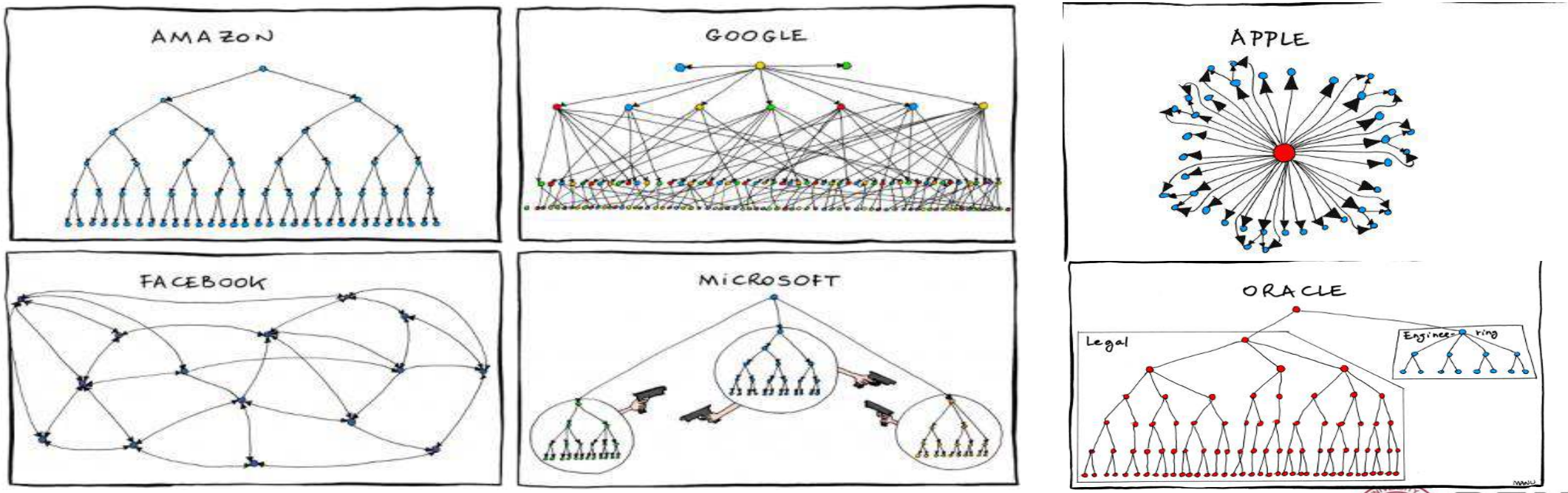
- **Information:** Anything that can be digitized (encoded as a stream of bits).
  - **Information goods:** *books, databases, magazines, movies, music, stock quotes, web pages, Weibo Trending Topics* (微博热搜)，*Short Videos* (短视频）, etc.

- Information is costly to *produce* by easy to *reproduce:*
  - *High fixed costs* and *low marginal costs*: first copy vs. additional copies
  - Implications:
    - **Pricing strategy:** Charge different prices for different versions of the same information good, e.g., advance on demand viewing (超前点播）
    - **Intellectual property rights protection:** Maximize the value, not the cost of protection，e.g., video trailers（视频预告）



# Diseconomies of Scale

- Reasons for **diseconomies of scale**
  - Coordination problems in large organizations
    e.g. management becomes over-stretched; cost out of control
  - Transportation costs tend to rise as production grows, due to handling expenses, insurance, security, and inventory costs



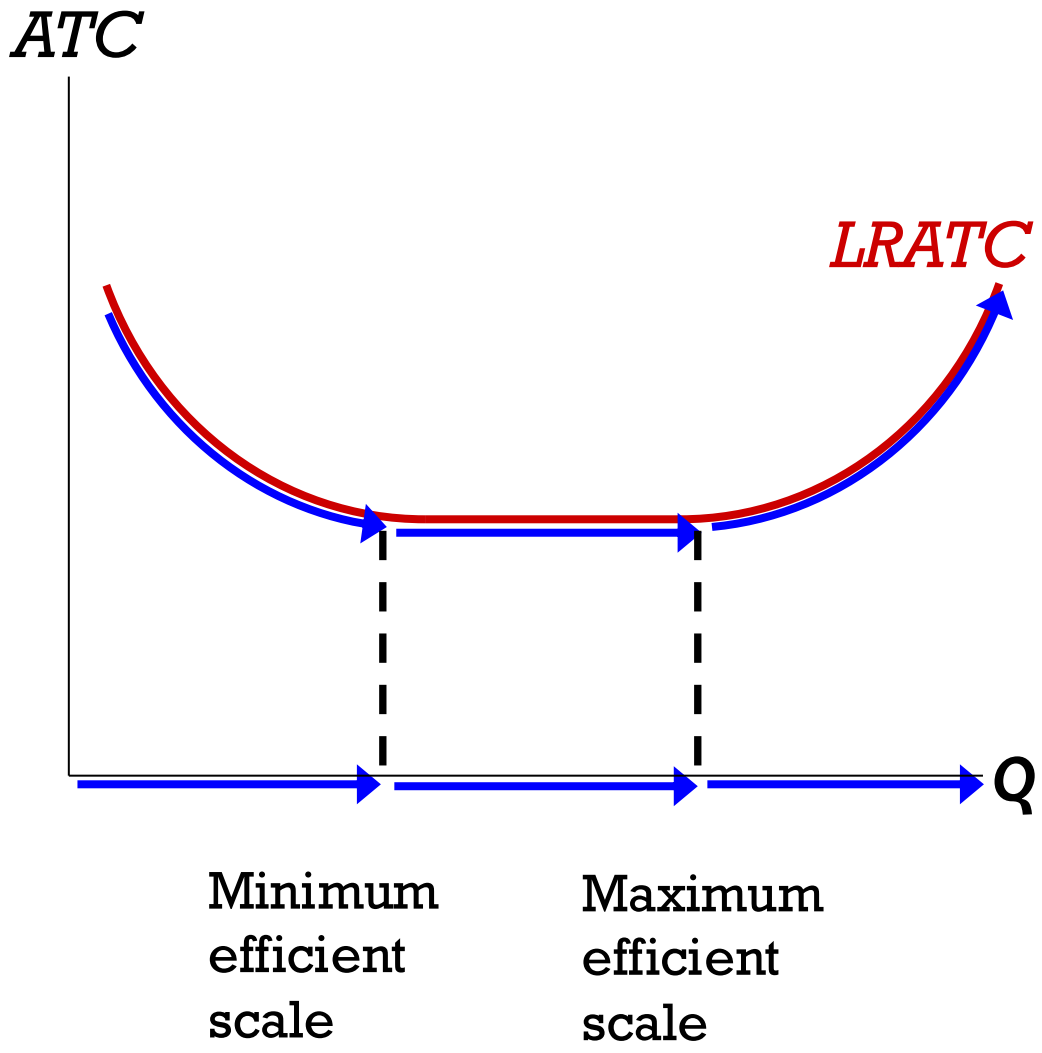
# Efficient Scales

- Flat section of the LAC displays **constant returns to scale**

  - The **minimum efficient scale (最小效率规模)** is the smallest scale at which minimum per unit costs are attained
    The start (left) point of the flat section

  - The **maximum efficient scale (最大效率规模)** is the largest scale before which unit costs begin to rise.
    The end (right) point of the flat section

# A Summary of ATC and Scale of Production



# Cost-reducing Strategies

# Strategy I: Learning by Doing （干中学）

**Learning curve**（学习曲线）: a line showing the relationship between cost and additional units of output

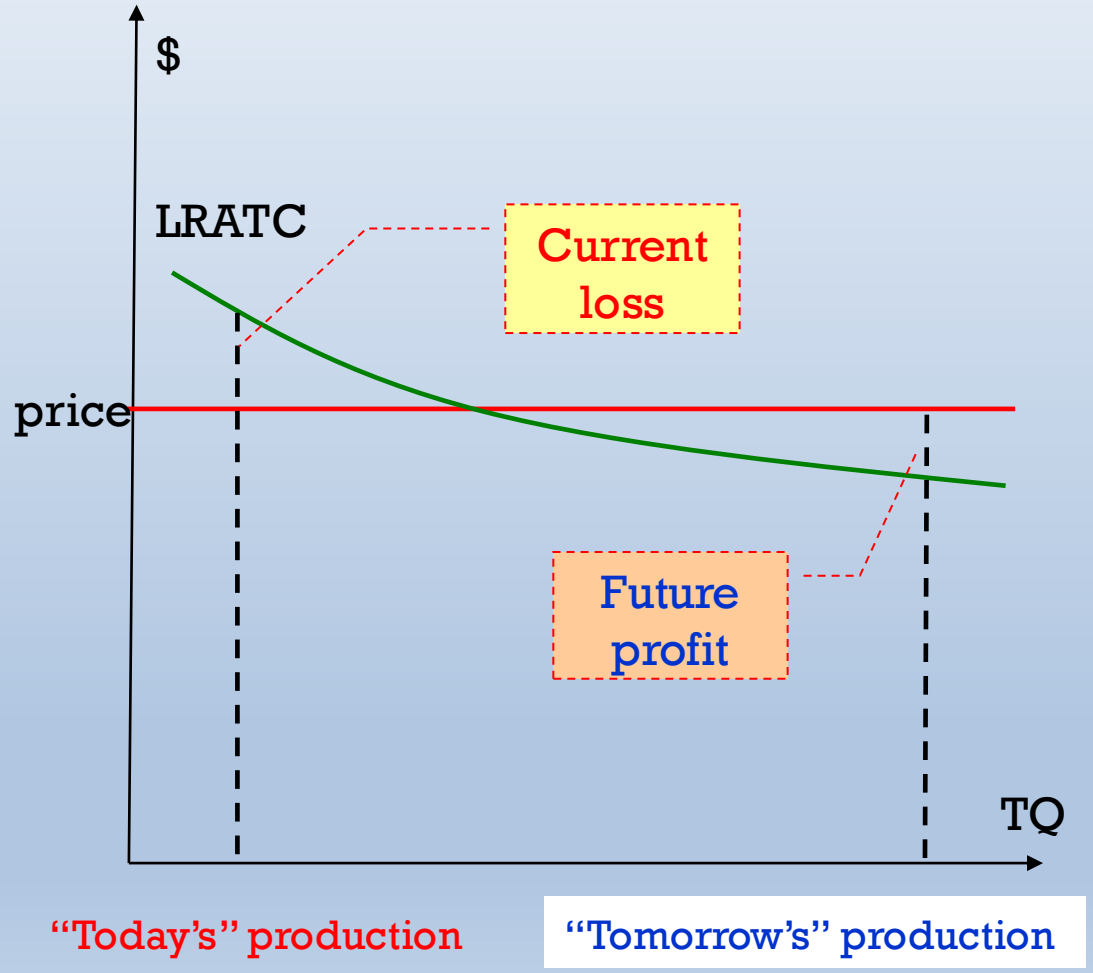
- Downward slope indicates additional cost per unit declines as the level of output increases because workers improve with practice (learn from experience)
- *Historical examples*: the take-off of manufacturing in some East Asian industrialized economies (South Korea, Chinese Taipei, etc)
  - Accumulating working experience to pulldown costs
  - Well known for the production of computer chips and consumer electronics

# Learning by Doing: Case Study of Snow Breweries

华润雪花啤酒(中国)有限公司
CHINA RESOURCES SNOW BREWERIES

## Four-Step Strategy:

1. Aggressive price-cutting measures
2. More orders
3. Accelerate production experience
4. Production cost becomes lower



# Strategy II: Economies of Scope（范围经济）

Reduction of a firm's unit cost by producing two or more goods or services jointly rather than separately

- A reputable *mineral company* is relatively cost-effective to expand into *soft drink* market

  (1) The company already has the knowledge, experience, and skills to produce and distribute.

  (2) More importantly, it has a good reputation among its customers.


农夫山泉
NONGFU SPRING
水溶C100
茶π
NFC
100%
农夫果园
30%混合果蔬

Coca-Cola


紅茶花伝
ROYAL ORANGE TEA
紅茶花伝
ROYAL BLACK TEA
紅茶花伝
ROYAL MILK TEA
純悦
美粒果
美汁源
果粒橙
普洱
淨茶含消茶
水动乐
纖綠茶
原萃
雪碧
纖維+
FANTA
POWERADE
爆锐
Coca-Cola
zero SUGAR
DASANI
PURIFIED WATER
fuze tea
AQUARIUS
Minute Maid
ORIGINAL
LOW PULP
COSTA
COFFEE
BODYARMOR
Strawberry Banana
PEACE TEA
NON GMO
FRESH TASTE GUARANTEED
Simply Peach
fairlife
Schweppes
Spicy Ginger Beer
LACTOSE FREE
POWERADE
SPORTS DRINK

# Strategy III: Supply Chain Management

**Supply chain management** (SCM，供应链管理): efforts by a firm to improve efficiencies through each link of a firm's supply chain from supplier to customer, reducing coordination costs

- "Make versus Buy" dilemma: vertical integration（垂直一体化） versus outsourcing（外包）: which one is more cost effective?
  - Empirical studies find that both exert inverted "U" shaped effect on firm performance, i.e.
  - A certain degree of implementation of both strategies are beneficial to a firm
  - Beyond an optimal point, further integration or outsourcing becomes harmful
  - "China + 1 strategy" in the decoupling era (HBR, 2021)

# Strategy IV: Process Innovation

- ***Product Innovation*** and ***Process Innovation*** are the two common types of innovation:
  - **Product Innovation**: invent a new or significantly improved product, e.g., *iphone, Cybertruck*
  - **Process Innovation**: implement new or significantly improved production, payment or organizational method, e.g., *Ford' Moving Assembly line, Toyata Production System (TPS)/Just-in-Time (JIT) Manufacturing, Taylorism (Scientific Management)*
- *Process innovation* includes many intangible/managerial innovation that helps improve production efficiency

**Human-Order Integration (人单合一) – Haier**

The Chinese appliance giant Haier dismantled its traditional hierarchical structure. It restructured into micro-enterprises (MEs)—small, self-managed teams that directly interface with customers. These MEs operate like internal startups; they are empowered to make decisions and are compensated based on the value they create for users. This links employee rewards directly to market outcomes, fostering extreme agility and customer-centricity .

# Summary

- Types of Costs in Managerial Decisions
  - **Short-Run:** Fixed costs, Sunk costs, Variable costs, Total costs
  - **Long-Run:** All costs are variable, no fixed costs

- Cost-reducing strategies:
  - Learning-by-doing
  - Economies of Scope
  - Supply Chain Management
  - Process innovations